

Embedded systems Exp 13

Program:

```
#include <reg51.h>
```

```
void main(void)
```

```
{
```

```
    unsigned int i;
```

```
    unsigned int array[5] = {0x1111, 0x2222, 0x8888, 0x4444, 0xABCD};
```

```
    unsigned long sum = 0;
```

```
    for (i = 0; i < 5; i++)
```

```
    {
```

```
        sum = sum + array[i];
```

```
    }
```

```
    P0 = (unsigned char)(sum & 0xFF);
```

```
    P1 = (unsigned char)((sum >> 8) & 0xFF);
```

```
    P2 = (unsigned char)((sum >> 16) & 0xFF);
```

```
    while (1);
```

```
}
```

output

•

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Registers Disassembly

Register Value

Regs

r0 0x11
r1 0x08
r2 0x00
r3 0x02
r4 0x00
r5 0x00
r6 0x08
r7 0x02

Sys

a 0x00
b 0x08
sp 0x15
sp_max 0x17
dptr 0x08
pc

exp13.c STARTUP.AS1

10 {
11 sum = sum + array[i];
12 }
13
14 P0 = (unsigned char)(sum & 0xFF);
15 P1 = (unsigned char)((sum >> 8) & 0xFF);
16 P2 = (unsigned char)((sum >> 16) & 0xFF);
17
18 while (1);
19 }

Command

WS 1, array[4]
WS 1, array[5], 0x0A
WS 1, array[5]
WS 1, array[5]

ASM ASSIGN BreakDisable BreakEnable BreakKill BreakList BreakSet BreakAccess COVERAGE COVTOFILE

Call Stack - Locals

Name Location/Value Type

MAIN C:0x0909
i 0x0005 uint

Memory 1 Call Stack - Locals

Watch 1

Name	Value	Type
i	0x0005	uint
sum	0x0001ABCC	ulong
array[0]	0x1111	uint
array[1]	0x2222	uint
array[2]	0x8888	uint
array[3]	0x4444	uint
array[4]	0xABCD	uint

<Enter expression>